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Probability and Statistics

Assignment Project Report

Determining the Impact of 3D Printer Parameter Adjustments on Print Quality - Accuracy - Durability

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1 Introduction

1.1 Project Overview

Additive manufacturing, commonly known as 3D printing, has revolutionized modern manufacturing processes by enabling the creation of complex geometries with unprecedented design freedom. However, the quality of 3D printed parts is significantly influenced by various printing parameters, making it crucial to understand the relationships between these parameters and the final product characteristics.

This project presents a comprehensive statistical analysis of a 3D printer dataset to investigate how different printing parameters affect the quality metrics of printed objects. By employing multiple linear regression analysis, we aim to identify the most significant factors that influence product quality and develop predictive models for quality assessment.

1.2 Research Objectives

The primary objectives of this research are:

- To analyze the relationship between 3D printing parameters and product quality metrics
- To identify the most influential parameters affecting surface roughness, tensile strength, and elongation
- To develop multiple linear regression models for predicting quality outcomes
- To provide data-driven insights for optimizing 3D printing processes
- To validate model assumptions and assess the reliability of our statistical findings

1.3 Dataset Description

The dataset used in this analysis was sourced from Kaggle and originates from research conducted by Selcuk University. It contains comprehensive measurements from 3D printing experiments with the following characteristics:

- **Sample size**: 1,000 observations
- **Input parameters**: 9 printing variables including layer height, wall thickness, infill density, infill pattern, nozzle temperature, bed temperature, print speed, material type, and fan speed
- **Output variables**: 3 quality metrics measuring roughness, tensile strength, and elongation
- **Data type**: Continuous numerical values for most variables, with categorical variables for infill pattern and material type

This dataset provides a robust foundation for statistical modeling, offering sufficient sample size and comprehensive coverage of key printing parameters that affect product quality.

1.4 Report Structure

This report is organized into seven chapters that systematically present our analysis:

- **Chapter 1 - Introduction**: Project overview, objectives, and dataset description
- **Chapter 2 - Data Cleaning and Preprocessing**: Data loading, quality assessment, and cleaning procedures
- **Chapter 3 - Descriptive Statistics**: Summary statistics, data visualization, and exploratory analysis
- **Chapter 4 - Methodology**: Statistical methods, multiple linear regression theory, and analytical approach
- **Chapter 5 - Results and Analysis**: Model development, diagnostic testing, and interpretation of findings
- **Chapter 6 - Conclusion**: Summary of results, practical implications, and recommendations
- **Chapter 7 - Data Source and References**: Dataset information and bibliographic references

Each chapter builds upon the previous one, creating a comprehensive narrative that guides the reader through our analytical process from data preparation to final conclusions.

1.5 Data Cleaning and Preprocessing

This chapter presents the systematic approach used to load, assess, and clean the 3D printer dataset. The data preprocessing pipeline ensures data quality and prepares the dataset for subsequent statistical analysis.

1.5.1 Data Loading

The first step involves loading the dataset and examining its basic structure. The following R code demonstrates the data loading process:

Listing 1.1: Data Loading and Initial Exploration

```r
# Load necessary libraries
library(readr)
library(dplyr)

# Load the 3D printer dataset
data <- read_csv("/home/kari/prob-stat-pj/3dprintdata/data.csv")

# Display basic information about the dataset
str(data)
head(data, 10)
colnames(data)
summary(data)
```

The dataset contains 50 observations with 12 variables, including 9 input parameters (layer_height, wall_thickness, infill_density, infill_pattern, nozzle_temperature, bed_temperature, print_speed, material, fan_speed) and 3 output quality metrics (roughness, tension_strength, elongation). The `str()` function reveals the data types, while `summary()` provides basic descriptive statistics for each variable.

Output:

```
Dimensions: 50 rows x 12 columns
Variables: layer_height, wall_thickness, infill_density, infill_pattern,
           nozzle_temperature, bed_temperature, print_speed, material,
           fan_speed, roughness, tension_strenght, elongation
```

1.5.2 Data Quality Assessment

Before proceeding with analysis, it is essential to assess the quality of the dataset by checking for missing values, duplicate records, and potential outliers.

Listing 1.2: Missing Values and Duplicate Detection

Missing Values and Duplicates

```r
# Check for missing values
missing_values <- sapply(data, function(x) sum(is.na(x)))
print(missing_values)
cat("Total missing values:", sum(missing_values), "\n")

# Check for duplicate rows
duplicate_count <- sum(duplicated(data))
cat("Number of duplicate rows:", duplicate_count, "\n")
```

The quality assessment revealed that the dataset is complete with no missing values (0 missing values across all variables) and no duplicate records (0 duplicate rows), indicating high data quality.

Output:

```
Missing values: 0
Duplicate rows: 0
```

Outlier Detection Outliers were detected using the Interquartile Range (IQR) method, which identifies values that fall outside the range [Q1 - 1.5×IQR, Q3 + 1.5×IQR]:

Listing 1.3: Outlier Detection Using IQR Method

```
# Define continuous variables for outlier detection
continuous_vars <- c("layer_height", "nozzle_temperature", "bed_
    temperature",
                    "print_speed", "infill_density", "wall_thickness",
                    "fan_speed", "roughness", "tension_strenght", "
                       elongation")

# Outlier detection for each continuous variable
for(var in continuous_vars) {
  if(var %in% colnames(data)) {
    Q1 <- quantile(data[[var]], 0.25, na.rm = TRUE)
    Q3 <- quantile(data[[var]], 0.75, na.rm = TRUE)
    IQR_val <- Q3 - Q1
    lower_bound <- Q1 - 1.5 * IQR_val
    upper_bound <- Q3 + 1.5 * IQR_val

    outliers <- data[[var]] < lower_bound | data[[var]] > upper_bound
    outlier_count <- sum(outliers, na.rm = TRUE)

    cat(sprintf("%s: Outliers = %d\n", var, outlier_count))
  }
}
```

The IQR method systematically evaluates each continuous variable by calculating the first quartile (Q1), third quartile (Q3), and the interquartile range. Values falling outside the calculated bounds are flagged as potential outliers for further investigation.

1.5.3 Data Cleaning Process

The data cleaning process involves handling any identified issues and preparing the data for analysis through appropriate data type conversions.

Data Type Conversions Categorical variables require conversion to factor type for proper statistical analysis:

Listing 1.4: Data Type Conversions

```
# Create a copy of original data for cleaning
data_cleaned <- data

# Convert categorical variables to factors
if("infill_pattern" %in% colnames(data_cleaned)) {
  data_cleaned$infill_pattern <- as.factor(data_cleaned$infill_pattern)
```

```
  cat("Converted infill_pattern to factor\n")
}

if("material" %in% colnames(data_cleaned)) {
  data_cleaned$material <- as.factor(data_cleaned$material)
  cat("Converted material to factor\n")
}

# Verify the conversion
str(data_cleaned)
```

The `infill_pattern` variable contains two levels (grid and honeycomb), while the `material` variable has two levels (ABS and PLA). Converting these to factors ensures they are treated as categorical variables in subsequent statistical analyses rather than character strings.

Output:

```
Factor variables: infill_pattern, material
infill_pattern levels: grid (25), honeycomb (25)
material levels: abs (25), pla (25)
```

Data Validation After cleaning, it is important to validate the data ranges and distributions:

Listing 1.5: Data Range Validation

```
# Check data ranges for continuous variables
for(var in continuous_vars) {
  if(var %in% colnames(data_cleaned)) {
    cat(sprintf("%s: Min=%.3f, Max=%.3f, Range=%.3f\n",
                var, min(data_cleaned[[var]], na.rm=TRUE),
                max(data_cleaned[[var]], na.rm=TRUE),
                max(data_cleaned[[var]], na.rm=TRUE) - min(data_cleaned
                    [[var]], na.rm=TRUE)))
  }
}

# Check categorical variable distributions
print(table(data_cleaned$infill_pattern))
print(table(data_cleaned$material))
```

This validation step ensures that all variables have reasonable ranges and that categorical variables have balanced distributions, which is important for the reliability of subsequent statistical analyses.

1.5.4 Cleaned Dataset Summary

The data cleaning process resulted in a high-quality dataset ready for analysis:

Listing 1.6: Final Dataset Summary

```
# Compare original and cleaned datasets
cat("Original dataset:", nrow(data), "rows x", ncol(data), "columns\n")
cat("Cleaned dataset:", nrow(data_cleaned), "rows x", ncol(data_cleaned)
    , "columns\n")
cat("Rows removed:", nrow(data) - nrow(data_cleaned), "\n")

```

```
# Final summary statistics
summary(data_cleaned)

# Save cleaned dataset
write_csv(data_cleaned, "/home/kari/prob-stat-pj/3dprintdata/data_
    cleaned.csv")
```

Key Findings from Data Cleaning The data cleaning process revealed several important characteristics:

- **Data Completeness**: The dataset is complete with no missing values across all 50 observations and 12 variables
- **Data Integrity**: No duplicate records were found, ensuring each observation represents a unique experimental condition
- **Variable Types**: Successfully converted 2 categorical variables (infill_pattern and material) to factors with balanced distributions (25 observations each for grid/honeycomb patterns and ABS/PLA materials)
- **Data Quality**: All continuous variables show reasonable ranges consistent with typical 3D printing parameters and quality metrics
- **Final Dataset**: The cleaned dataset maintains the original 50 × 12 structure, indicating no data loss during the cleaning process

Summary Statistics of Cleaned Dataset:

```
 layer_height    wall_thickness   infill_density   infill_pattern
Min.   :0.020   Min.   : 1.00   Min.   :10.0   grid     :25
1st Qu.:0.060   1st Qu.: 3.00   1st Qu.:40.0   honeycomb:25
Median :0.100   Median : 5.00   Median :50.0
Mean   :0.106   Mean   : 5.22   Mean   :53.4
3rd Qu.:0.150   3rd Qu.: 7.00   3rd Qu.:80.0
Max.   :0.200   Max.   :10.00   Max.   :90.0

nozzle_temperature bed_temperature  print_speed   material    fan_speed
Min.   :200.0      Min.   :60      Min.   : 40   abs:25   Min.   :  0
1st Qu.:210.0      1st Qu.:65      1st Qu.: 40   pla:25   1st Qu.: 25
Median :220.0      Median :70      Median : 60            Median : 50
Mean   :221.5      Mean   :70      Mean   : 64            Mean   : 50
3rd Qu.:230.0      3rd Qu.:75      3rd Qu.: 60            3rd Qu.: 75
Max.   :250.0      Max.   :80      Max.   :120            Max.   :100

  roughness      tension_strenght   elongation
Min.   : 21.0   Min.   : 4.00   Min.   :0.400
1st Qu.: 92.0   1st Qu.:12.00   1st Qu.:1.100
Median :165.5   Median :19.00   Median :1.550
Mean   :170.6   Mean   :20.08   Mean   :1.672
3rd Qu.:239.2   3rd Qu.:27.00   3rd Qu.:2.175
Max.   :368.0   Max.   :37.00   Max.   :3.300
```

The cleaned dataset provides a solid foundation for descriptive statistics and subsequent multiple linear regression analysis, with all variables properly formatted and validated for statistical modeling.

2 Descriptive Statistics

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References

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